

The Feynman Lectures on Physics, Volume II

Problem Set 7

Due 2022.6.12

You are encouraged to discuss the problem sets amongst your peers but the final product you turn in must be done on your own and presented in your own words and calculations. The bonus problems are optional, but I encourage you to do them.

1 The Magnetism of matter (4pt)

A charged particle of mass m moves in a plane perpendicular to a uniform magnetic field $\mathbf{B}$.

- (a) Show that if $\mathbf{B}$ changes slowly in time, the magnetic moment $\boldsymbol{\mu}$ produced by the orbital motion remains constant.
- (b) What, precisely, do we mean by "slowly" in part a?

2 Paramagnetism and Magnetic Resonance (4pt)

Derive an equation for quantum mechanical paramagnetism for particles of spin one, following the derivation in Chapter 35 of Vol. II for spin one-half.

3 Ferromagnetism (4pt)

A uniformly magnetized sphere has a total magnetic moment of $\mu = 4\pi a^3 M/3$, where a is the radius, and M the magnetization.

- (a) Calculate the equivalent surface current per unit length K which can replace this sphere as far as external effects are concerned.
- (b) Show that this current distribution has the same total magnetic moment.

4 Bonus: The Principle of Equivalence (4pt)

Please to try use the equivalent principle to derive the equation of motion for a particle in gravitational field.

Hint: The equivalent principle tells us that in the free fall frame ξ^α , the equation of motion is that of a straight line in spacetime. That is

$$\frac{d^2\xi^\alpha}{d\tau^2} = 0 \tag{1}$$

Now suppose that we use any other coordinate system x^μ , which may be a Cartesian coordinate system at rest in the laboratory. And x^μ is a function of ξ^α . Using the chain rule of derivative, we can write down the general equation of motion in arbitrary frame x^μ .